WeForming

Middleware
User Manual

CONSORTIUM PARTNERS

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1 Purpose and scope

This manual describes how to use the **WeForming Data Space Onboarding and Governance Platform** to:
(i) onboard a new participant into the WeForming Data Space, (ii) track onboarding status and issued artefacts and (iii) perform governance actions (participant admission, VC issuance and revocation) when operating with administrator privileges.

The manual covers two operational roles:

- **Participant (Onboarding Console):** initiates onboarding, generates/handles identity bootstrap artefacts, tracks requests, views credentials and accesses audit information.
- **Administrator (Governance Console):** performs all participant actions plus privileged governance workflows, including admission decisions and credential lifecycle operations.

This manual documents the participant (onboarding) workflow first, and then only the additional governance screens (since the governance console is a superset).

2 Roles, access and prerequisites

2.1 Roles

- **Participant:** can create onboarding requests and manage self-service views ("My requests", "My credentials", "My audit trail")
- **Administrator:** can access governance navigation items and execute privileged actions (admission control, credential lifecycle operations)

2.2 Prerequisites

Before starting, ensure you have:

- Access credentials to the platform (participant or administrator account).
- Required organisational metadata (organisation name, contact details, endpoints) as applicable.
- The target connector deployment environment available (to consume onboarding outputs later)

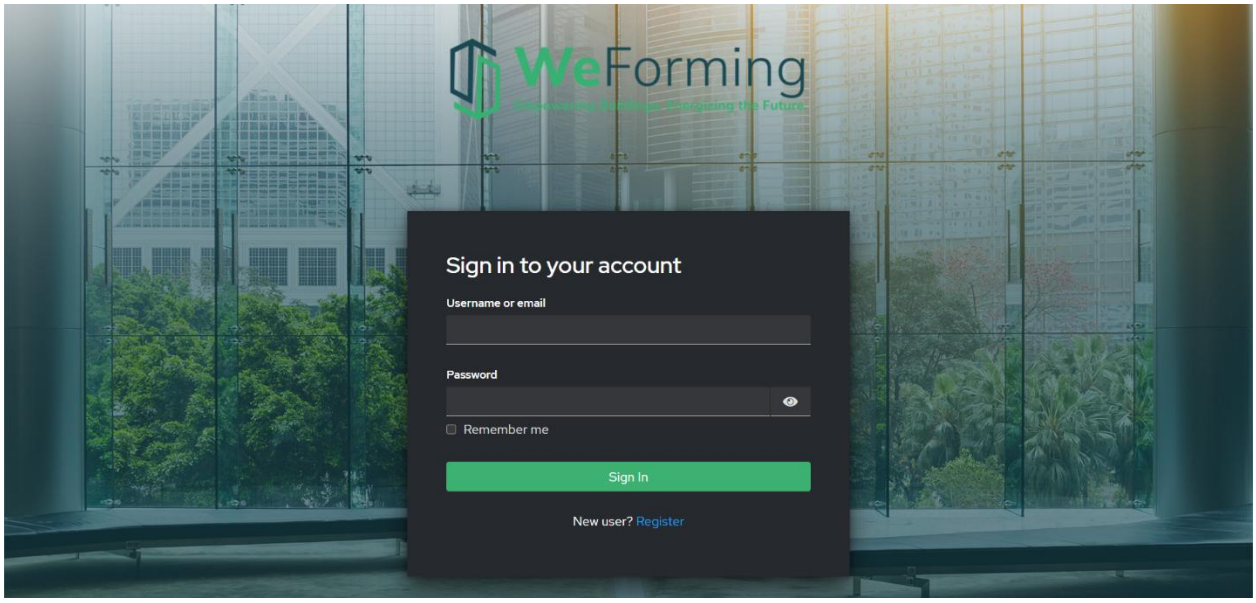


Figure 1: Login Screen

3 Platform overview and navigation (common for both roles)

After login, the platform shows a left-side navigation menu. The **participant** sees onboarding-related entries; the **administrator** additionally sees **Governance** entries.

3.1 Common navigation (Participant view)

- **Onboarding**
- **Onboarding status**
    - My requests
    - My credentials
    - My audit trail

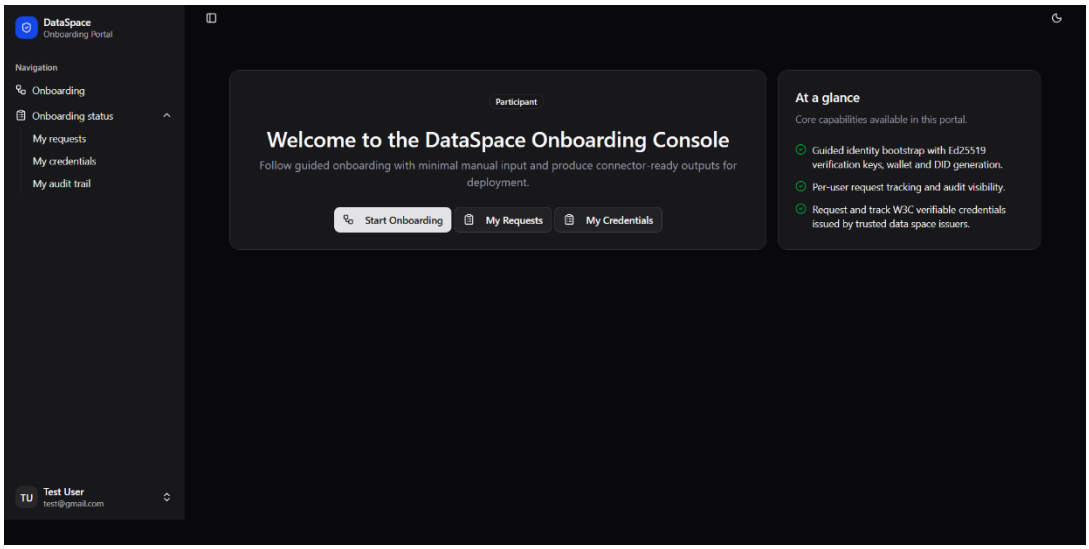


Figure 2: Onboarding Platform (Participant View)

3.2 Additional navigation (Administrator view)

- **Governance**
    - Incoming requests
    - Resolved requests
    - VC issuance
    - VC revocation

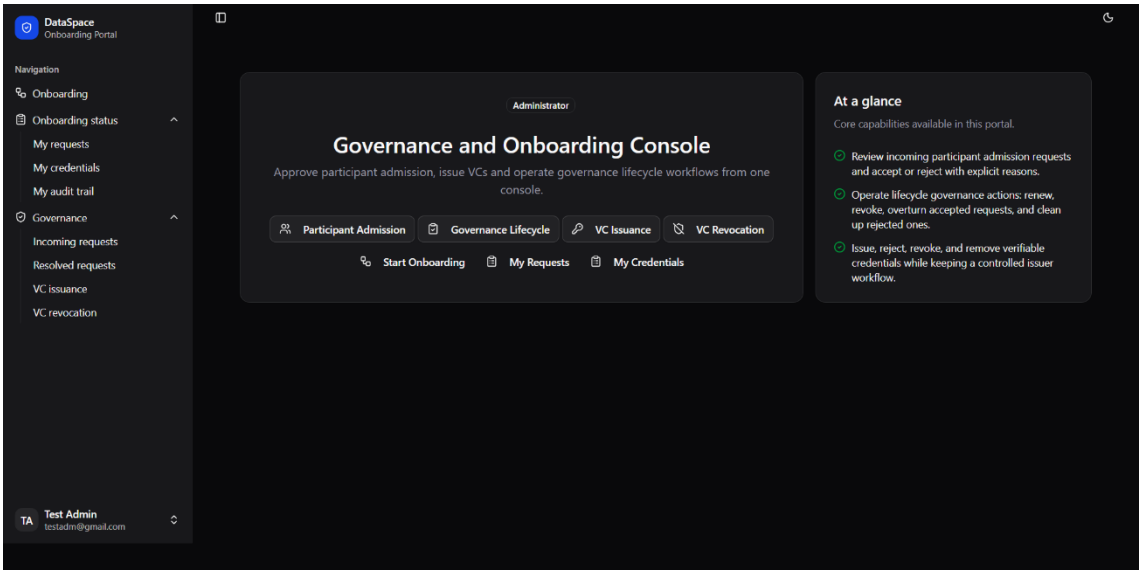


Figure 3: Governance Platform (Administrator View)

Part A - Participant Manual (Onboarding Console UI)

4 Onboarding wizard (Step-by-step)

The onboarding process is implemented as a **6-step guided wizard**. The wizard is designed to minimise manual handling while keeping sensitive materials under participant control.

***Security note**: Wallet creation and Ed25519 key pair generation are performed **client-side in the user's browser**. Private key material is **not stored by the platform backend**. Participants are responsible for secure storage of private artefacts and for providing them to the subsequent connector deployment procedure.*

4.1 Step 1 of 6 - Organisation Details

In this step, the participant provides a minimal organisation profile used by trusted issuers/attestors and governance operators to evaluate the admission request and VC eligibility. Typical fields include:

- **Organisation name** (legal name)
- **Intended participation role** (Provider / Consumer / Prosumer)
- **Intended use in the Data Space** (short description)
- **Contact details** (email, website, jurisdiction)
- Optional **additional information** (supporting information for validation)

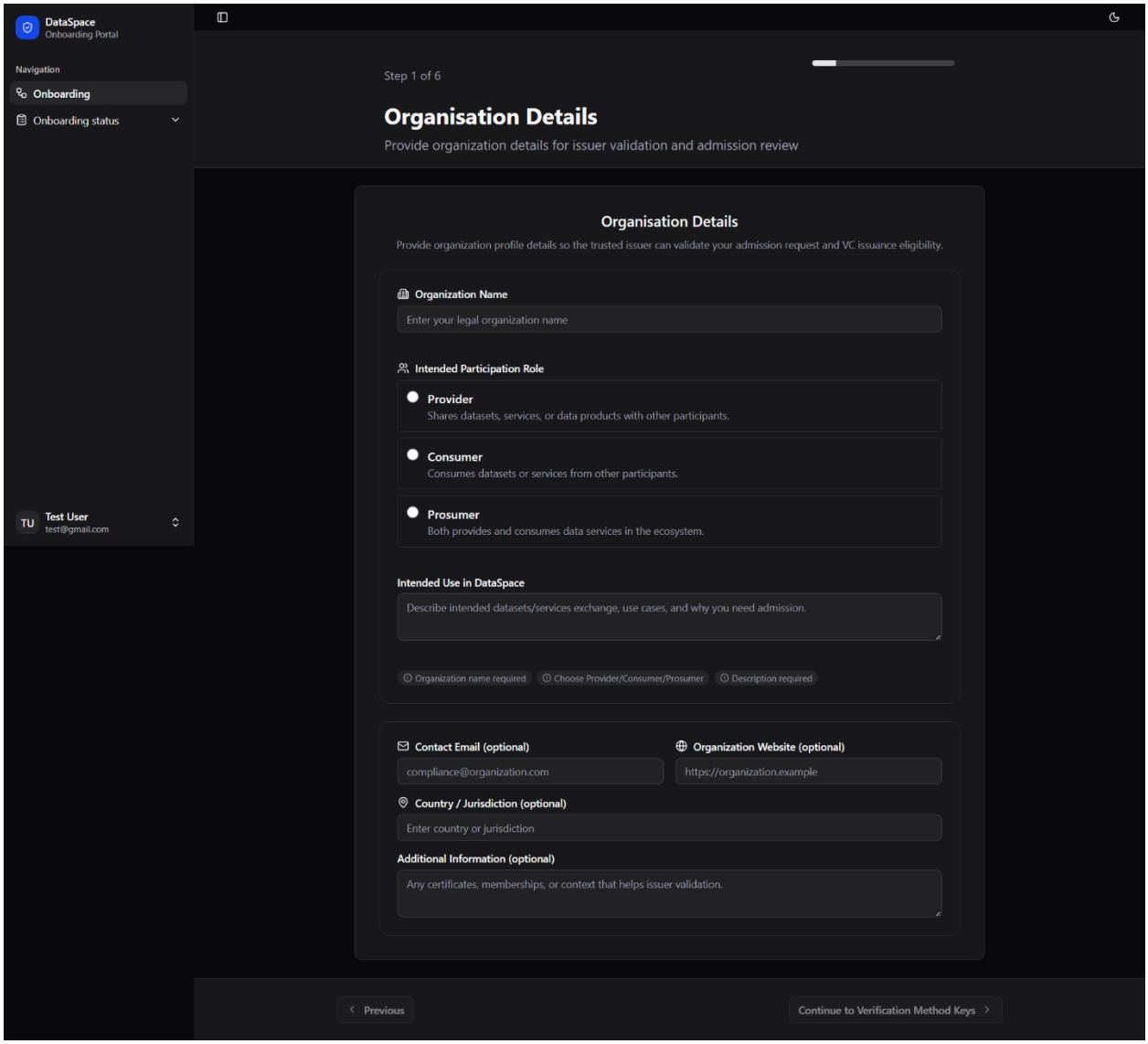


Figure 4: Onboarding Wizard - Step 1

4.2 Step 2 of 6 - DID Verification Method Keys (Ed25519)

The participant generates (or uploads) an **Ed25519 key pair** used as DID verification method keys for signing and authentication. The platform:

- generates keys client-side, or accepts an uploaded PEM key pair
- publishes the **public key** into the participant's DID document
- keeps the **private key** under participant control (download/export)

Operational options typically include:

- Generate automatically

- Upload existing Ed25519 key pair (PEM)
- Download public key / private key
- View fingerprint (SHA-256) for integrity checking

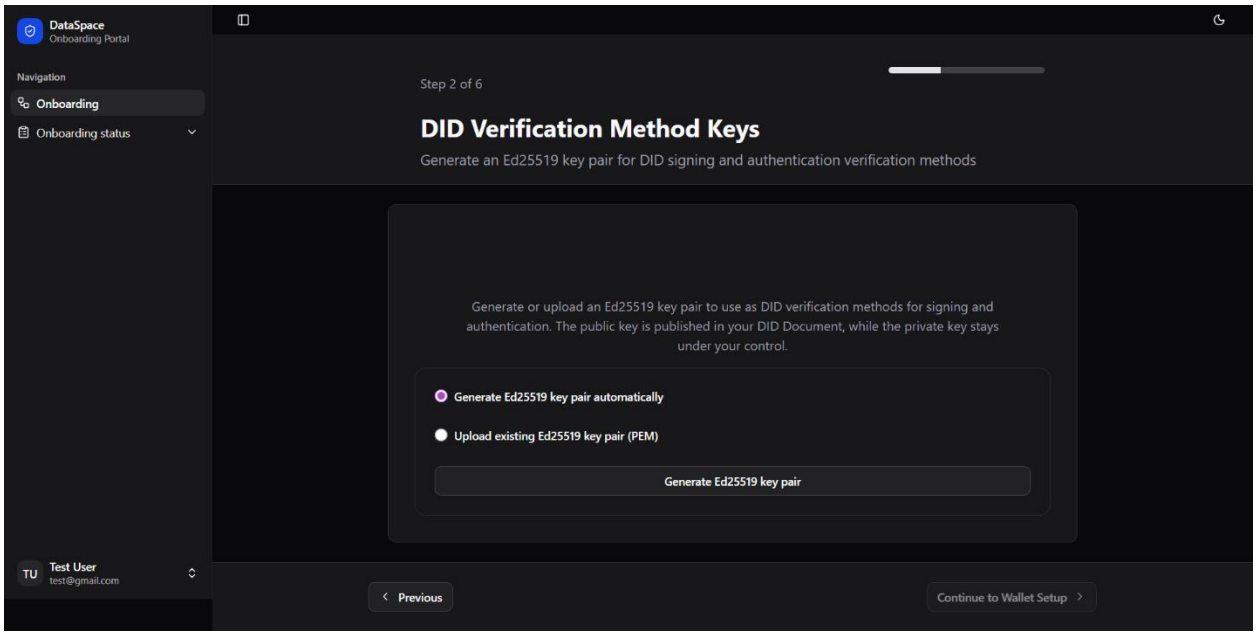


Figure 5: Onboarding Wizard - Step 2 (Before key generation)

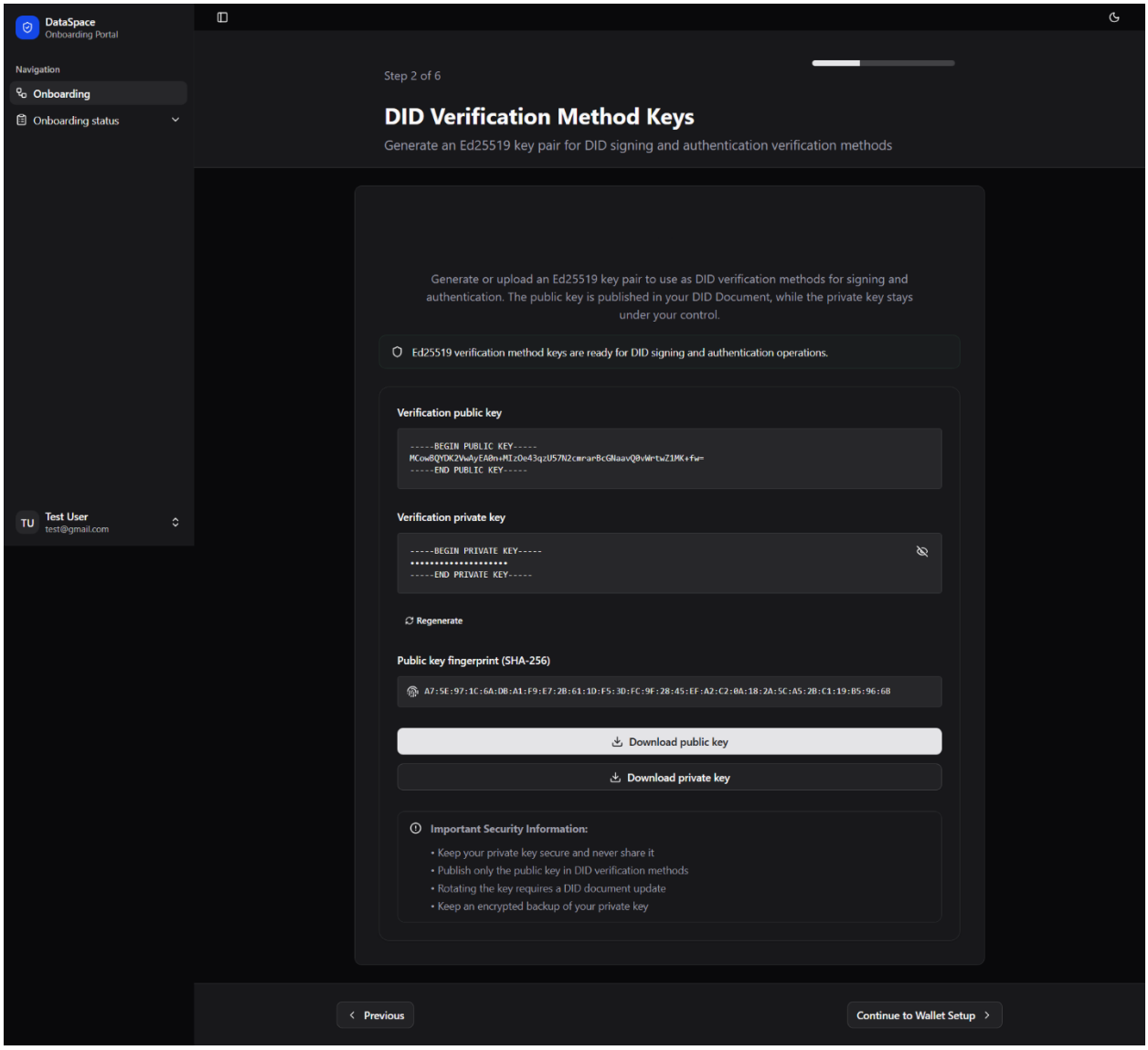


Figure 6: Onboarding Wizard - Step 2 (After key generation)

4.3 Step 3 of 6 - Wallet Setup (Blockchain wallet)

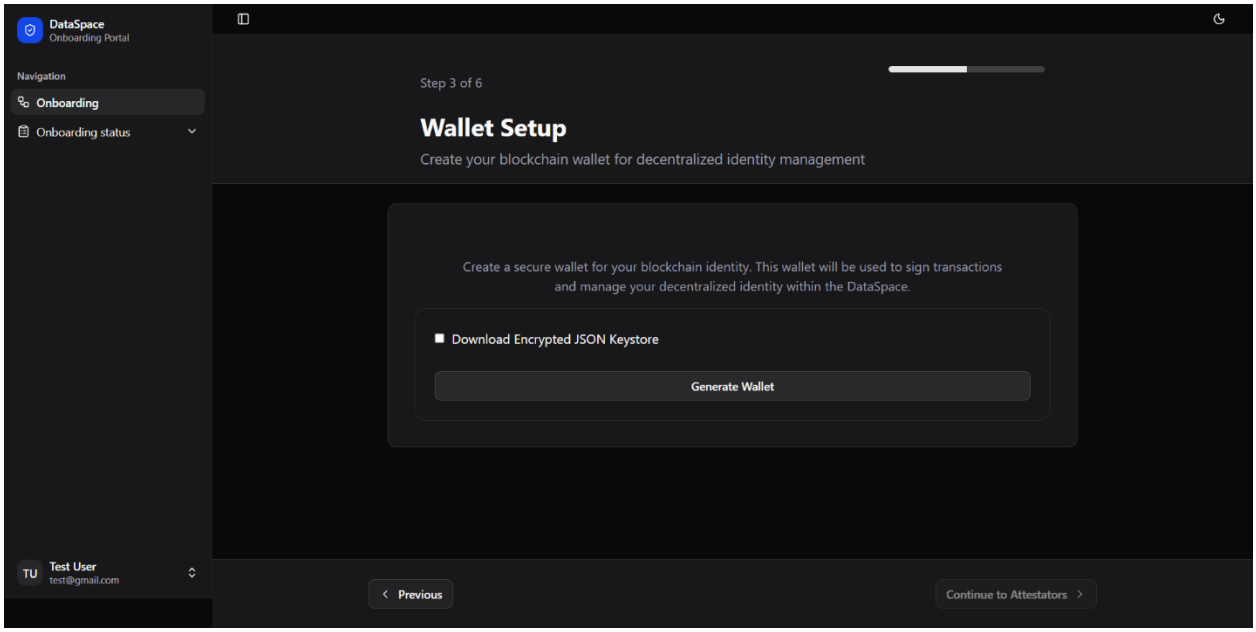
In this step, the participant creates a blockchain wallet that will be used to sign identity-related transactions and anchor the decentralised identity within the WeForming trust infrastructure. **Wallet creation is performed client-side in the user's browser** to ensure that sensitive key material remains under participant control.

The platform:

- generates the wallet and displays the resulting **Wallet Address**;
- allows export of the **Wallet Private Key** for use in subsequent connector deployment/bootstrap activities;
- provides an option to download an **Encrypted JSON Keystore** for secure storage and later import.

The **Encrypted JSON Keystore** follows the standard **Web3 Secret Storage** format used in the Ethereum ecosystem. This enables encrypted wallet storage protected by a user-defined password and interoperability with compatible Ethereum tooling for recovery or deployment.

Reference (**Web3 Secret Storage specification**)



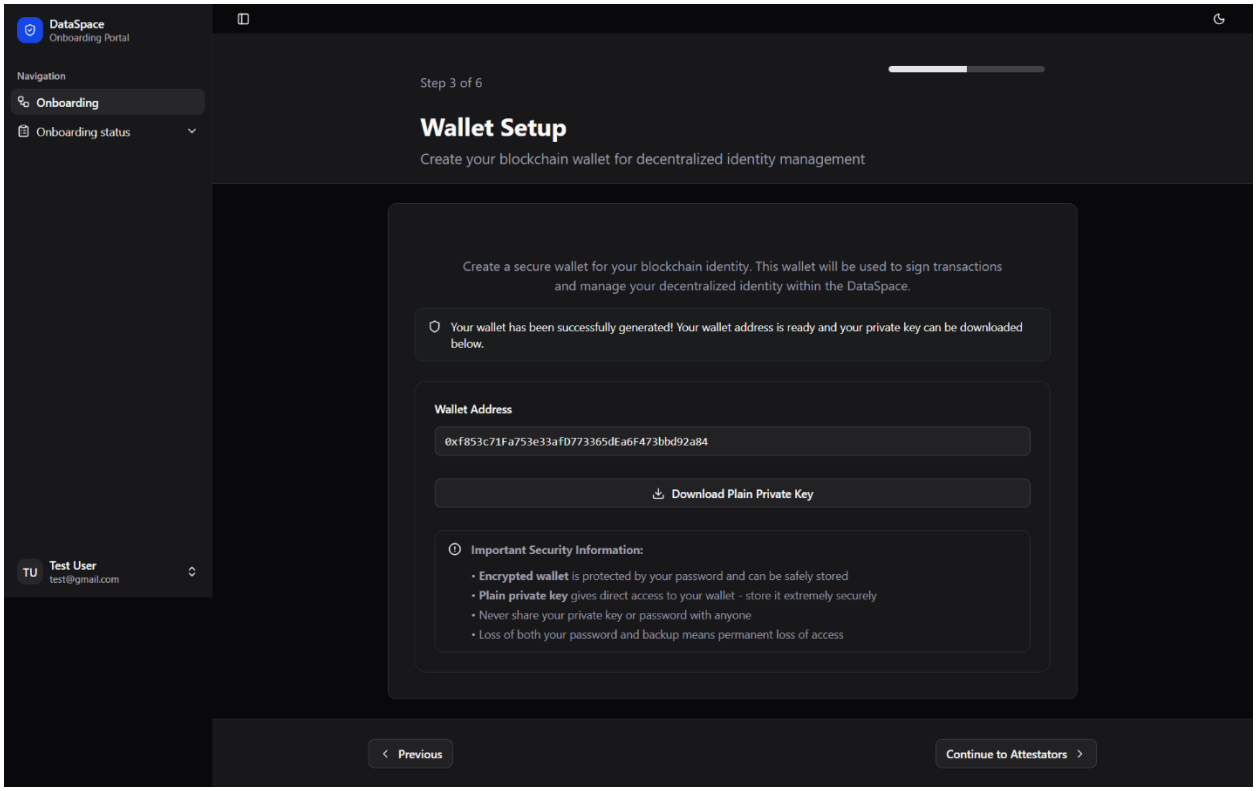


Figure 7: Onboarding Wizard - Step 3 (Unencrypted Wallet generation)

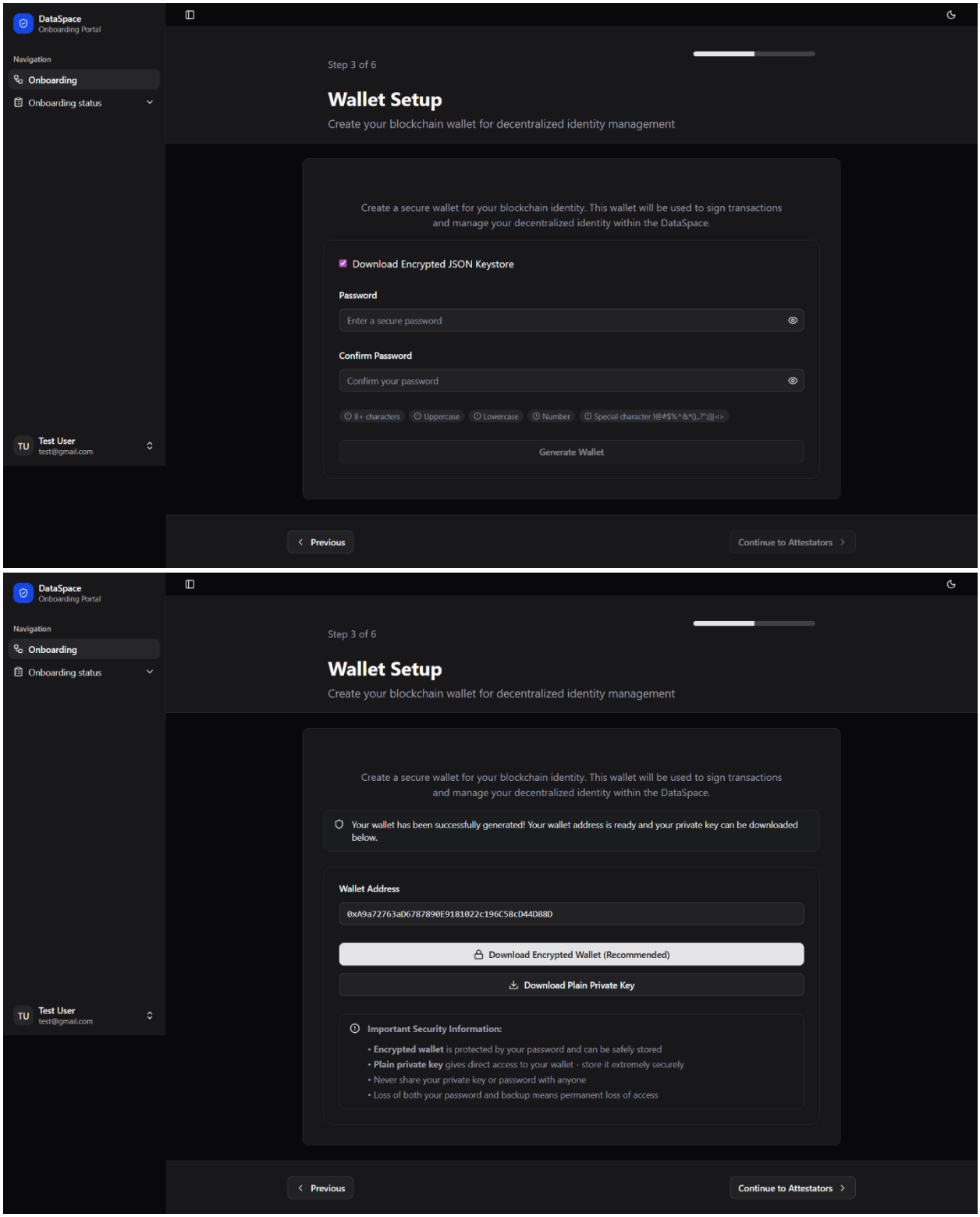


Figure 8: Onboarding Wizard - Step 3 (Encrypted Wallet generation)

4.4 Step 4 of 6 - Select Trusted Issuers / Attestators

The participant selects one or more **attestors / trusted issuers** that will verify the admission request and (where applicable) participate in credential issuance eligibility. This step establishes the governance pathway for approval in a permissioned environment.

The participant should select attestors aligned with their onboarding context (e.g., consortium operator, designated issuer organisations). Multiple attestors can be selected if broader verification coverage is required.

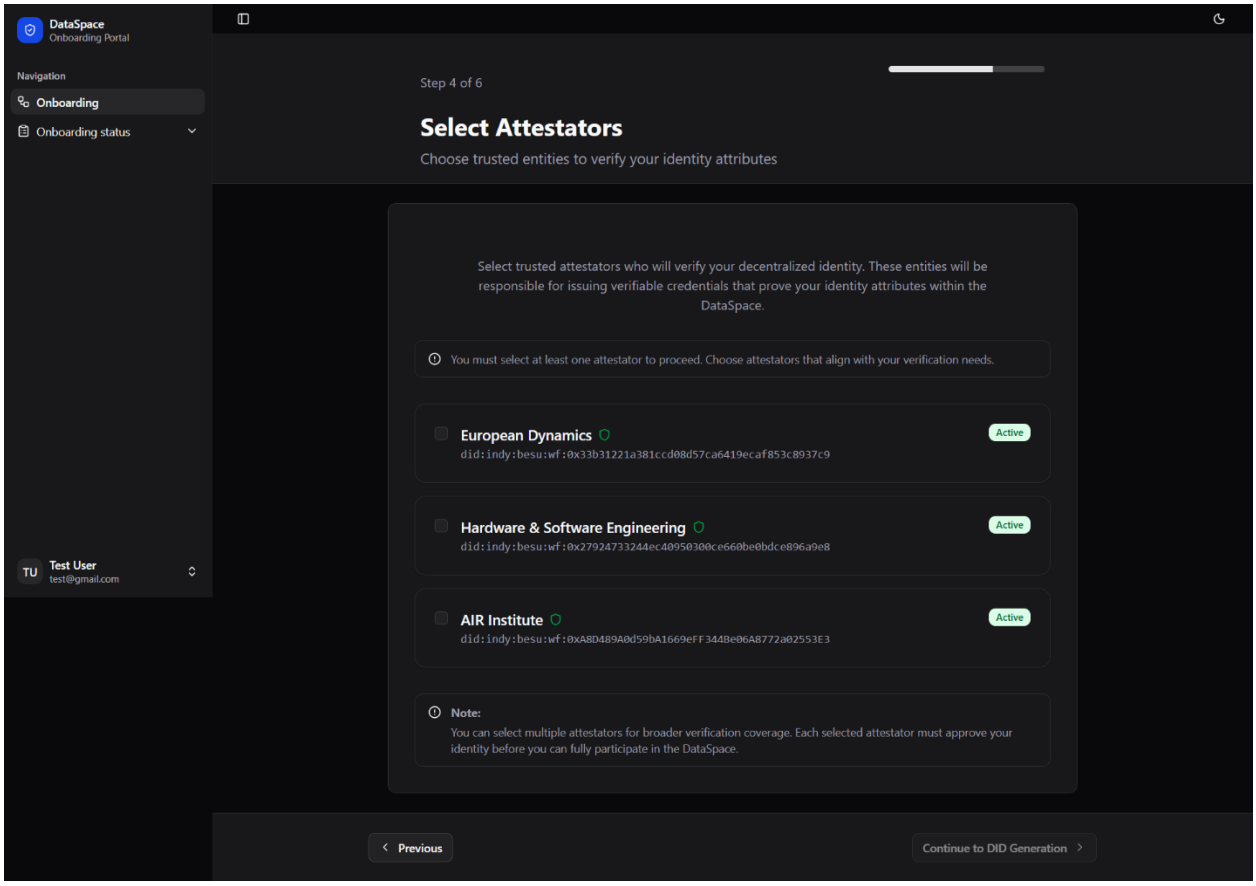


Figure 9: Onboarding Wizard - Step 4

4.5 Step 5 of 6 - DID Generation & Confirmation

The platform computes and presents the participant DID using the WeForming namespace convention:

did:indy:besu:wf:{WalletAddress}

The participant confirms the generated DID and verifies it matches the displayed wallet address. This step ensures the participant understands the binding between wallet control and the DID identifier.

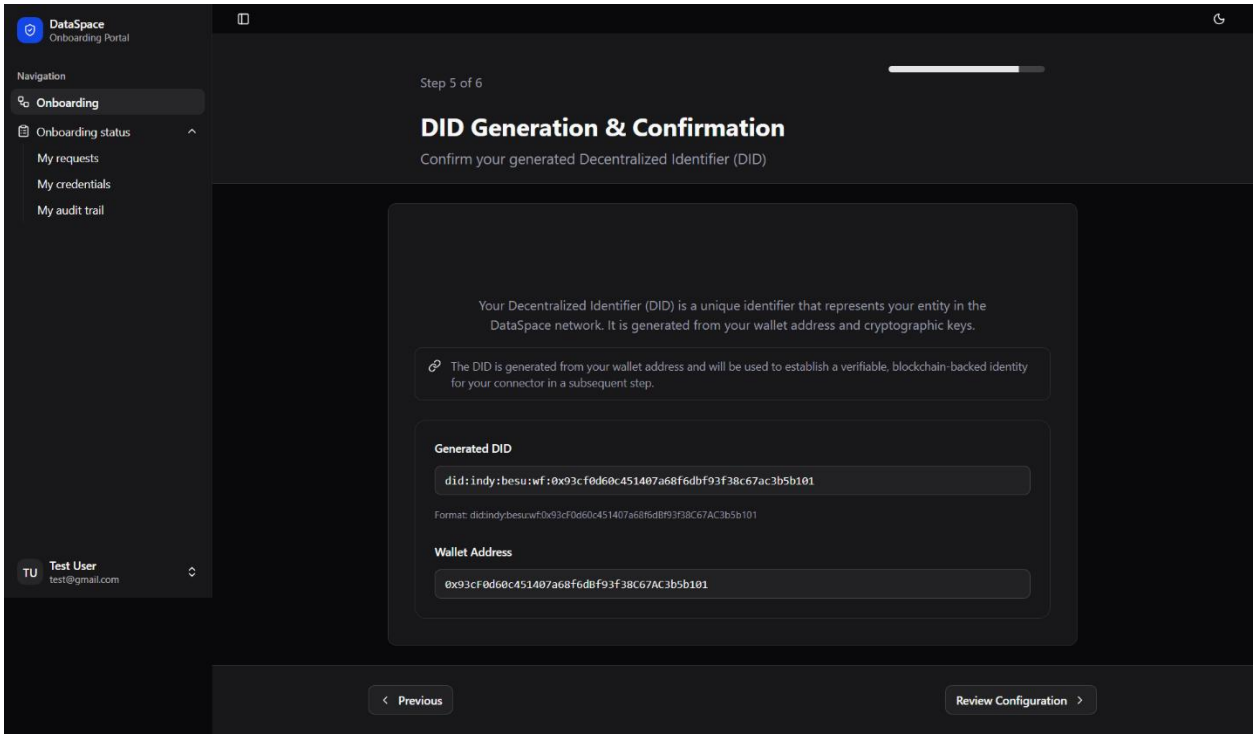


Figure 10: Onboarding Wizard - Step 5

4.6 Step 6 of 6 - Configuration Summary (Connector-ready outputs)

At the end of the wizard, the platform presents a summary of the onboarding request. Submitting the request triggers the governance validation workflow (performed by administrators in a permissioned setup).

Typical items shown include:

- **DID** (*did:indy:besu:wf:{WalletAddress}*)
- **Verification Method ID** (e.g., DID + #key-1)
- **Public key fingerprint** (SHA-256)
- **Wallet Address**
- **Attestors selected**
- **Blockchain network parameters** (pre-configured where applicable)
- **Download connector config** (downloads the summarised items)

The participant must capture and securely store the identity bootstrap artefacts that will be provided to the subsequent **Connector deployment** procedure. These artefacts are: (i) the **Wallet Address**, (ii) the **Wallet Private Key**, (iii) the participant **DID**, which follows the convention **did:indy:besu:wf:{WalletAddress}** and (iv) the **Ed25519 key pair** used for decentralised identity verification.

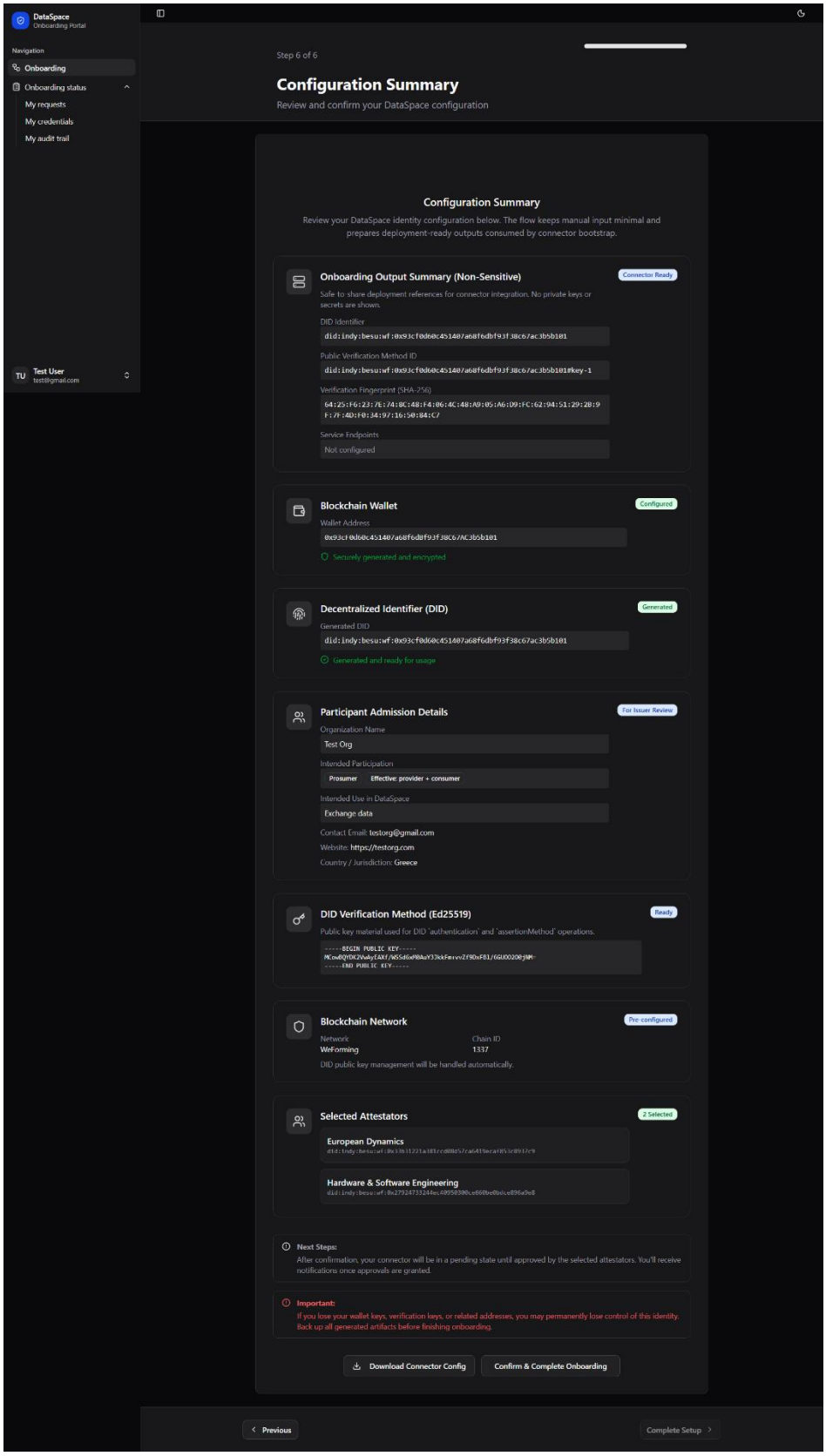


Figure 11: Onboarding Wizard - Step 6

5 Tracking onboarding progress and outcomes

5.1 My requests

Navigate to **Onboarding status → My requests** to track submitted onboarding requests. This view provides request state visibility (e.g., pending, approved, rejected) and timestamps, enabling the participant to understand where the onboarding stands in the governance process.

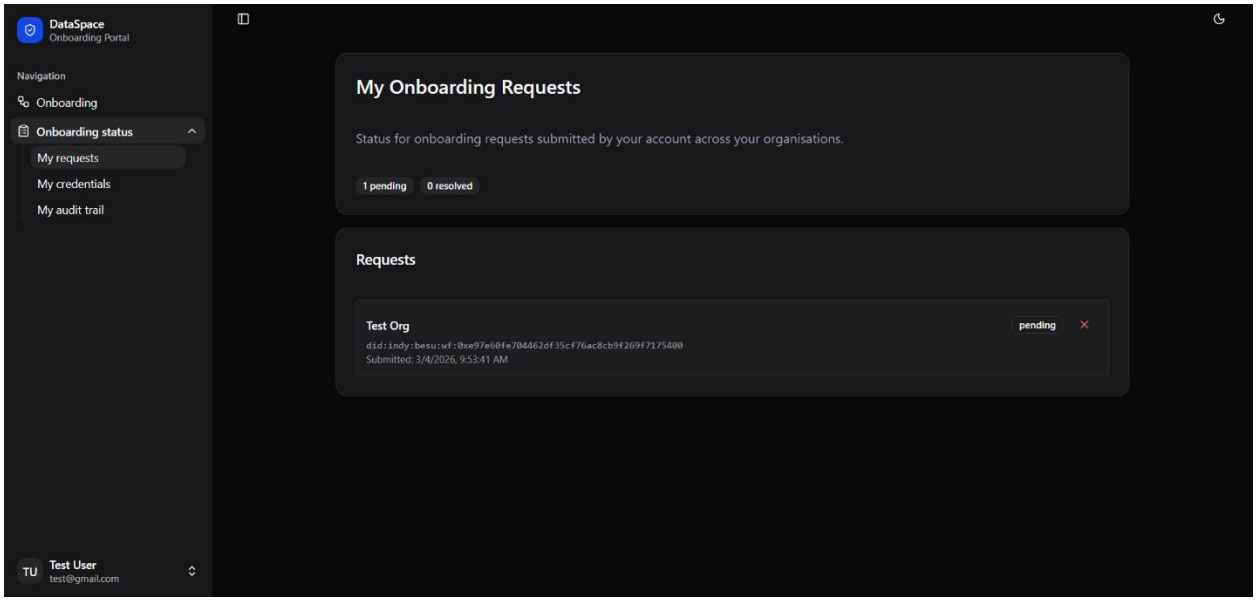


Figure 12: Onboarding Platform - My Requests Page

5.2 My credentials

Navigate to **Onboarding status → My credentials** to view credentials issued to the participant (e.g., membership/role VCs issued by trusted issuers under governance). This view enables the participant to confirm receipt of credentials required for authorisation and policy decisions during transactions. It also gives the participant the ability to request credentials from a trusted issuer/attestator (chosen from the Trusted Issuer Registry)

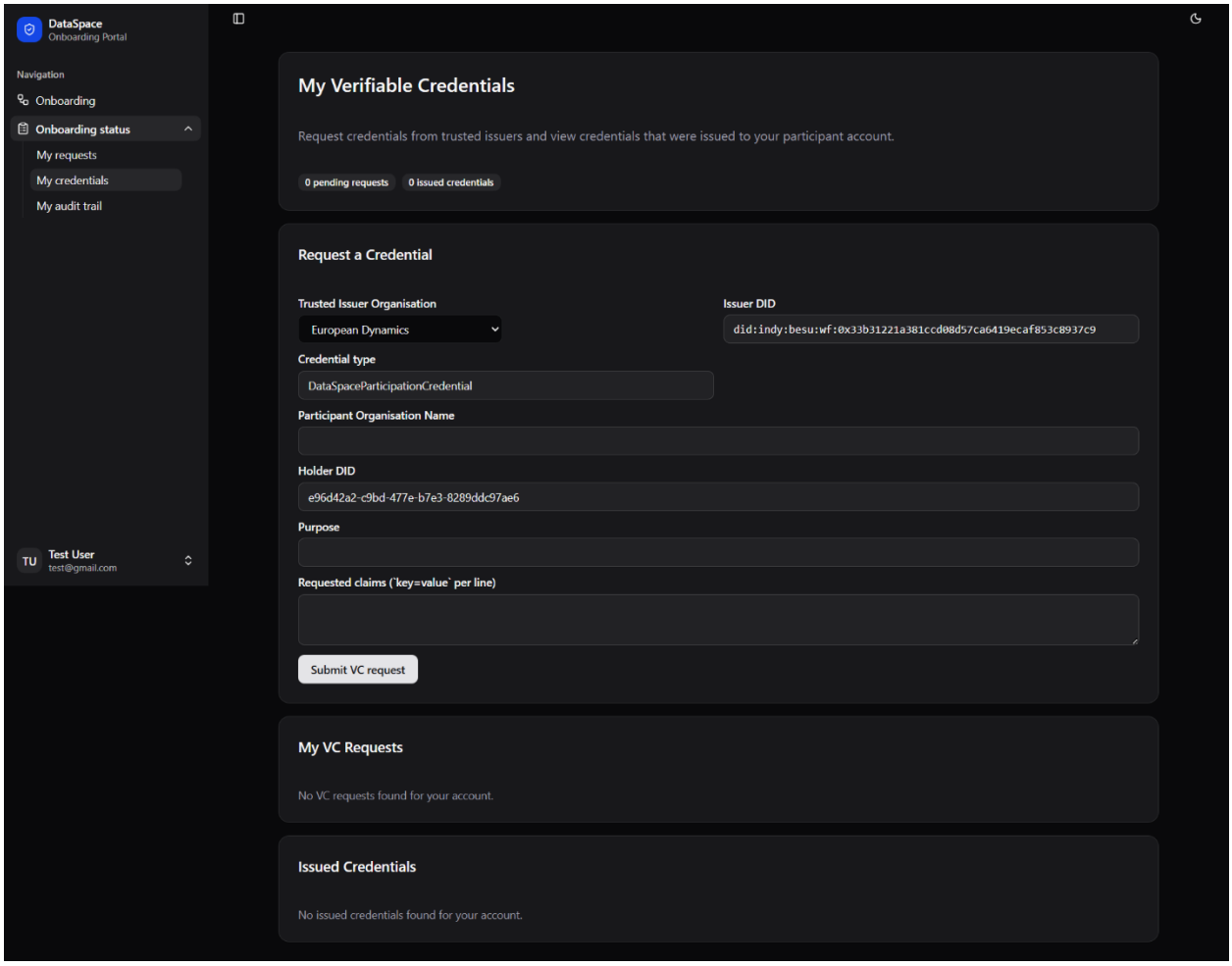


Figure 13: Onboarding Platform - My Credentials Page

5.3 My audit trail

Navigate to **Onboarding status → My audit trail** to view the audit-relevant record of actions and lifecycle events related to the participant's onboarding and governance interactions (e.g., submission, approvals, credential issuance).

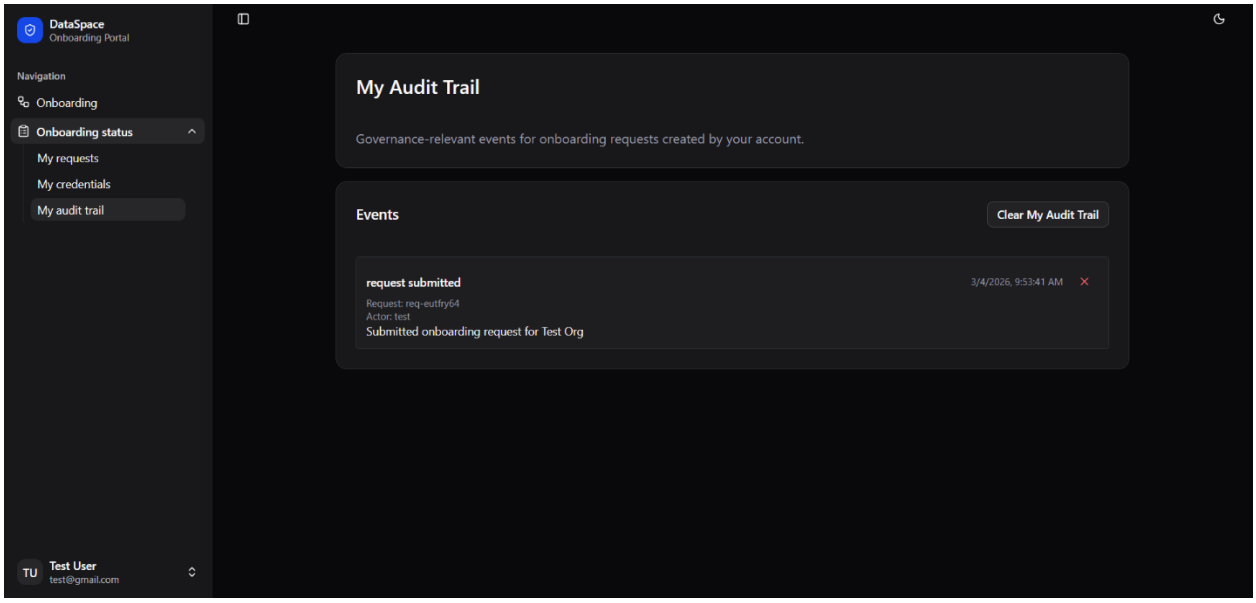


Figure 14: Onboarding Platform - My Audit Trail Page

6 Handover to connector deployment (what the participant must provide next)

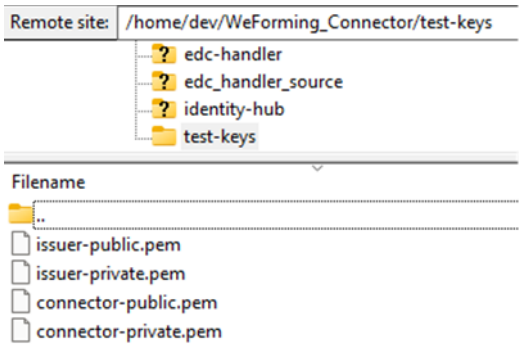
After completing onboarding, the participant must provide the following artefacts to the connector deployment process:

- **Wallet Address**
- **Wallet Private Key**
- **DID**, derived as **did:indy:besu:wf:{WalletAddress}**
- **Ed25519 key pair** (private key retained by participant; public key published into the DID document)

These artefacts are generated client-side; the participant is responsible for secure storage and controlled use during connector bootstrap.

To conclude the process, the connector owner shall upload the connector's configuration:

1. Add in the following Connector root (./WeForming_Connector/test-keys/..)the public and private key as delivered



2. Configure the did parameters on the connector file connector-config.properties @ ./WeForming_Connector/connector/config/connector-config.properties adding

```
edc.participant.id=did...
edc.iam.issuer.id=did...
edc.iam.sts.oauth.client.id=did...
edc.iam.sts.oauth.client.secret.alias=did...-sts-client-secret
edc.iam.sts.publickey.id=did...#key-1
```

Part B - Administrator Manual (Governance Console UI)

The administrator role has access to all participant features described above. The sections below describe **only the additional governance capabilities**, to avoid duplication.

7 Governance console overview (Administrator)

The Governance Console (Figure 3) provides a unified operational view to:

- evaluate onboarding requests
- approve or reject participant admission
- issue VCs and manage credential lifecycle (including revocation)
- maintain traceability for governance actions

8 Admission control and request lifecycle

8.1 Incoming requests

Navigate to **Governance → Incoming requests** to view requests awaiting review. Each request should include sufficient metadata to support evaluation, such as participant information, request timestamps, and onboarding artefact status.

Typical admin actions include:

- open request details
- perform compliance checks according to governance rules
- approve (admit participant) or reject (with reason)

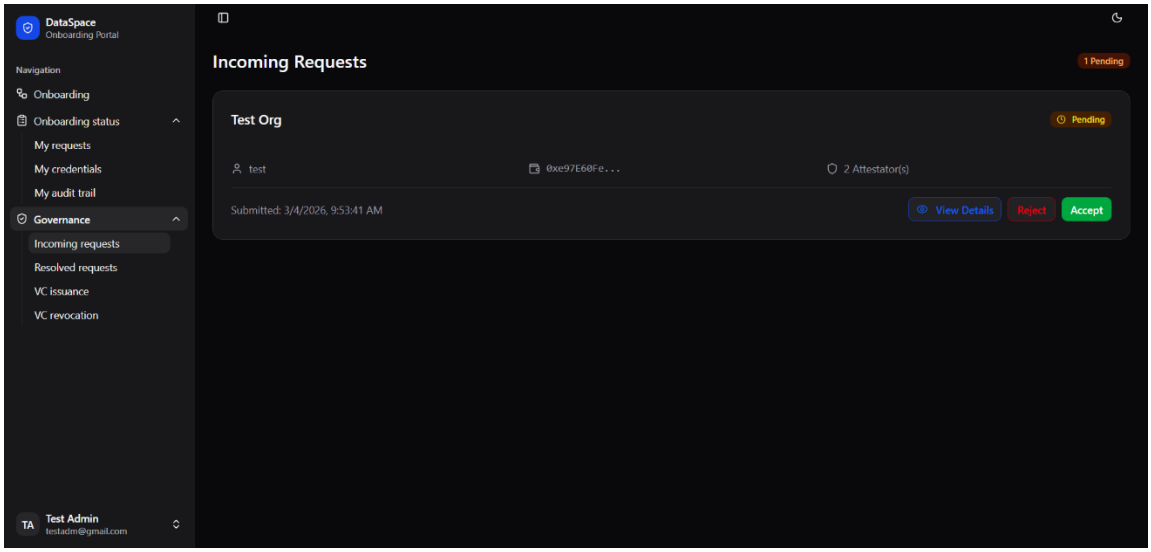
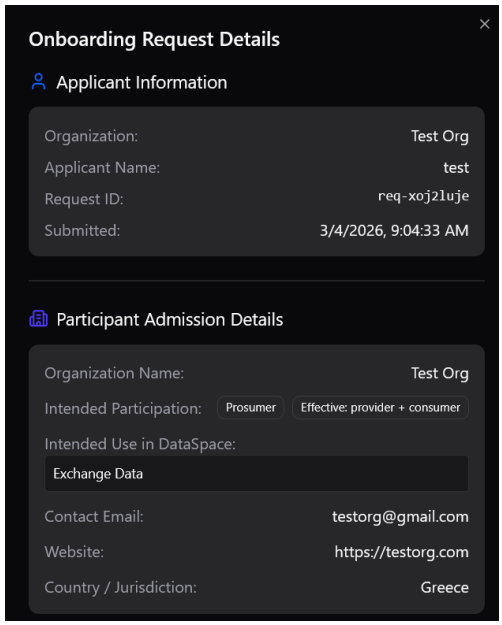


Figure 15: Governance Platform - Incoming Requests Page

8.2 Request detail and admission decision

Select a request to open its detailed view. The request detail view supports:

- verification of participant information
- validation of required onboarding artefacts
- execution of the admission decision



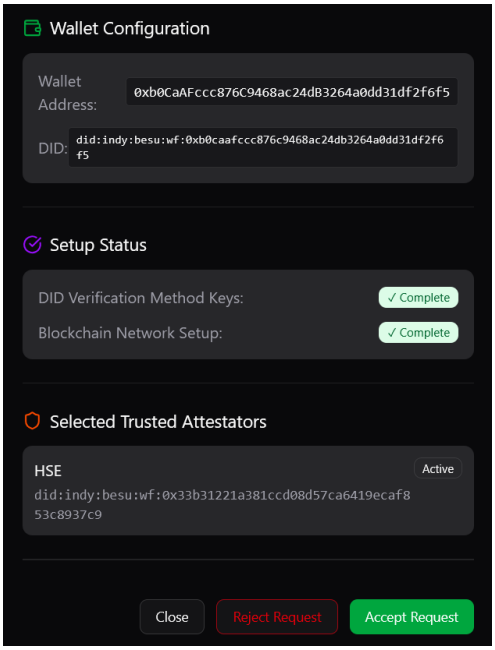


Figure 16: Governance Platform - Incoming Requests Page (Request Details View)

8.3 Resolved requests

Navigate to **Governance → Resolved requests** to review completed decisions and maintain operational traceability across the lifecycle of participant onboarding.

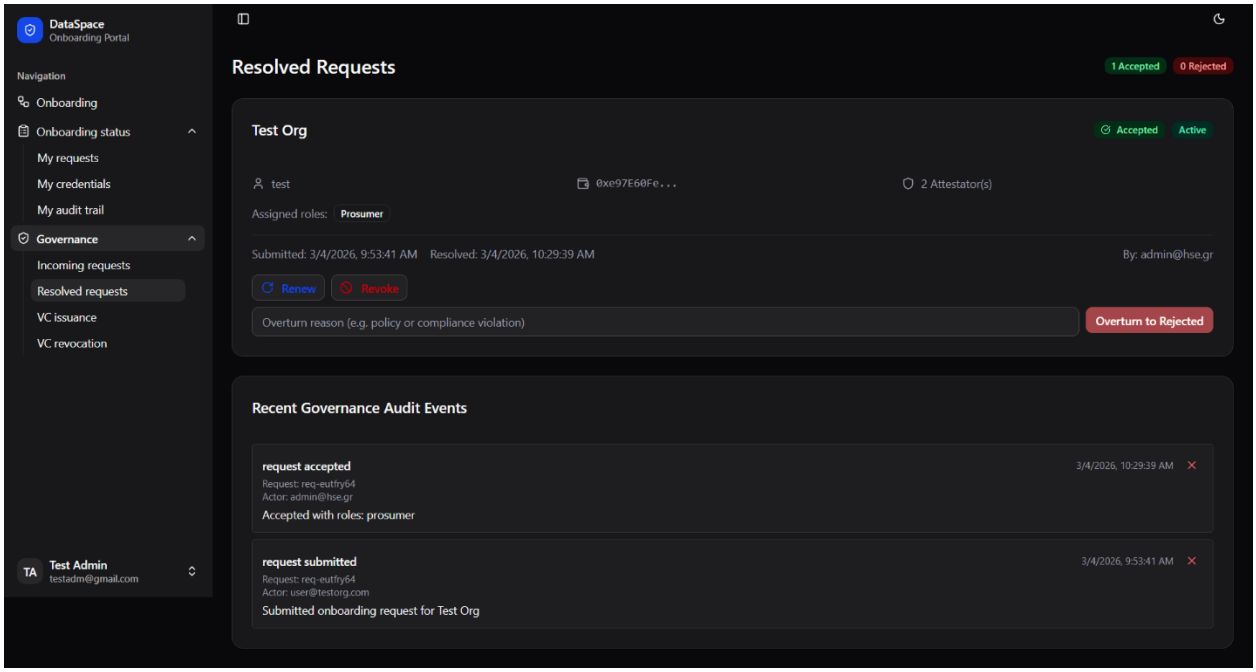


Figure 17: Governance Platform - Resolved Requests Page

9 Verifiable credential (VC) lifecycle operations

9.1 VC issuance (governance action)

Navigate to **Governance → VC issuance** to issue credentials expressing membership and/or roles for participants. This workflow is used to provide verifiable authorisation evidence that can be validated by WeForming components during transaction-time trust checks.

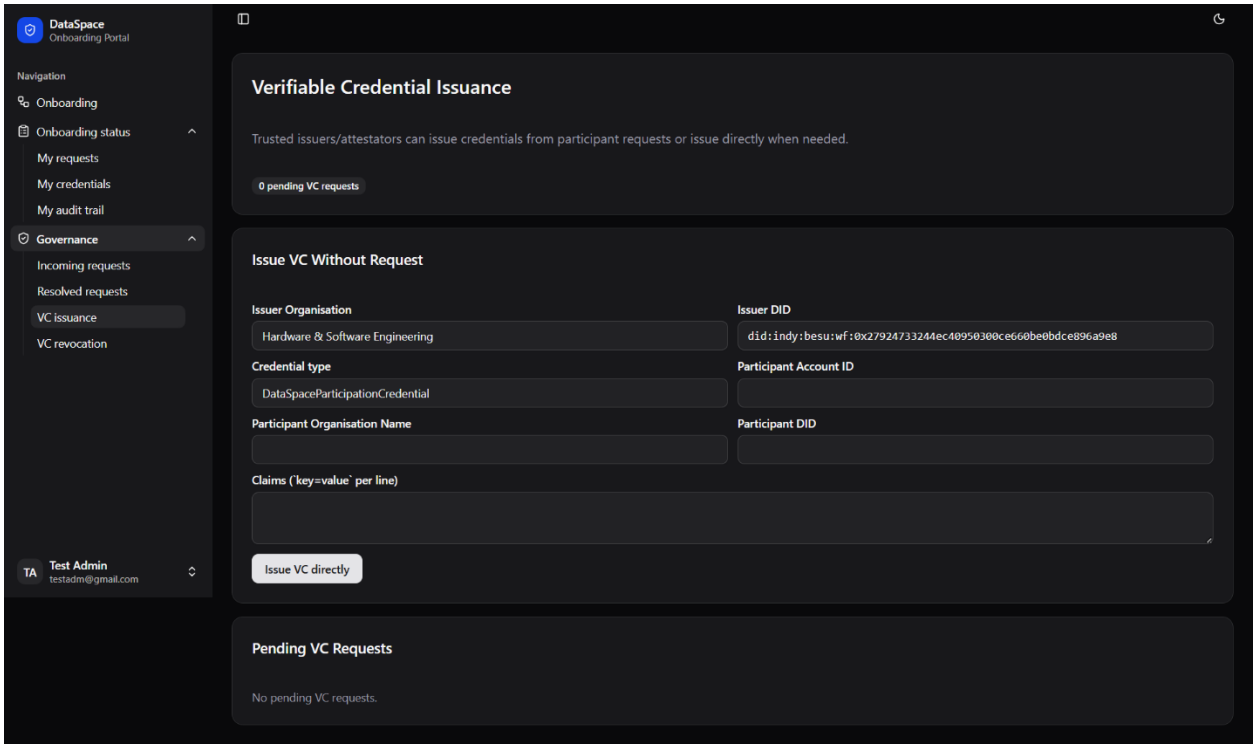


Figure 18: Governance Platform - VC Issuance Page

9.2 VC revocation (governance action)

Navigate to **Governance → VC revocation** to invalidate previously issued credentials when required (e.g., offboarding, role changes, compliance issues). Revocation ensures that trust decisions can reflect updated governance state, rather than relying on outdated authorisation evidence.

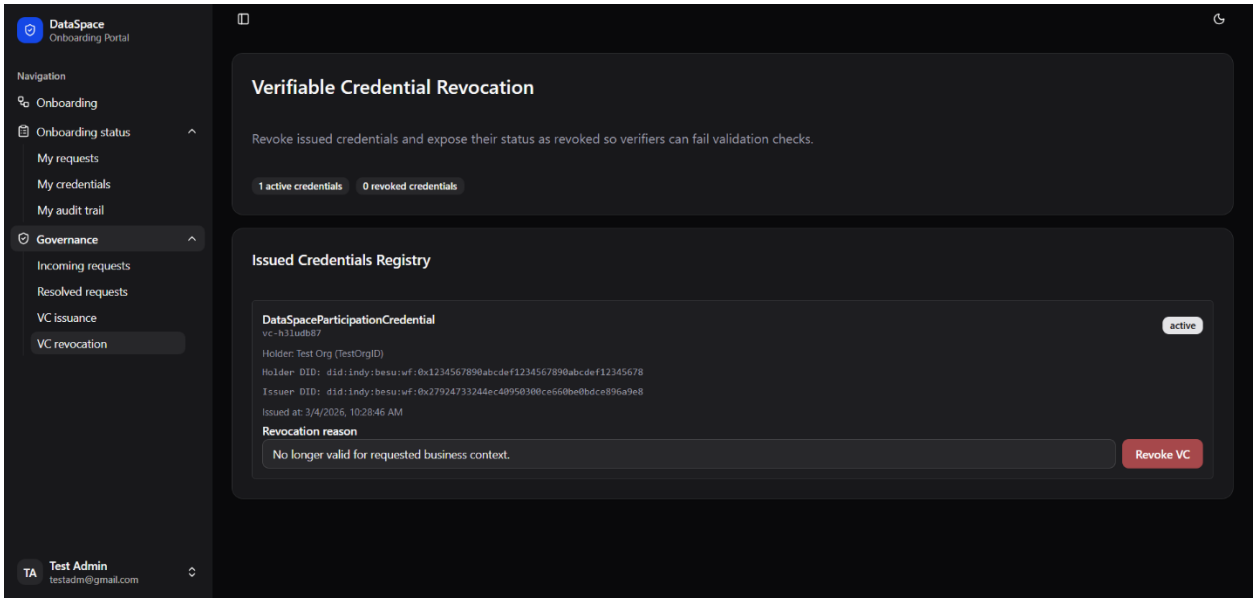


Figure 19: Governance Platform - VC Revocation Page

10 Governance operational guidance

- Apply consistent decision criteria for admission approvals/rejections to ensure non-discriminatory governance enforcement aligned with consortium rules.
- Treat governance actions as security-sensitive operations; ensure admin accounts follow strong authentication and least privilege.
- Ensure revocation decisions are traceable and, where applicable, communicated to affected participants through the platform's request/credential views.

11 Relationship with other components of the WeForming project

The onboarding and governance platform produces identity and governance outputs intended for consumption by:

- Connector deployment/bootstrap procedures (participant-side)
- Runtime policy decisions of the Middleware (transaction-time DID/VC verification)